Design History — Knotted Coil Designer

One place to see how the design evolved: every version, who asked for it and why, what changed, which checks were added, and what the checks found, failures included. Detailed rationale is in `DECISIONS.md`; Reza's requests, quoted, are in `docs/REQUEST_LOG.md`; the current specification is `Design.md`. Git hashes refer to this repository.

All work was done on 2026-10-20 unless a date is shown.

Version	Trigger	Main change	Checks added (pre-registered)	Outcome
v1.0 (git 39a4e83)	Reza: knotted receive loops, bloom, automatic decoupling, helmet, wire loops, up to 8 lobes	Torus-knot elements T(p,q); bloom = lobe amplitude (symmetric shape); Neumann-formula M and L; two-stage decoupling solver; browser app	V1 Maxwell coaxial M; V2 ring self-L; V3 coplanar-circle null; V4 convergence; V5 symmetry; V6 topology under bloom	Run 1: V1, V2, V4, V5 PASS; V3 NOT EVALUABLE (Roemer 1990 unread); **V6 FAIL** (bloom floor below 0.02, a point-count error, and a test-harness bug). Run 2: V6 FAIL only below the physical floor
(no version)	Reza: in-plane rotation as a lever	Rotation modes (free / locked / fixed) exposed	—	Free rotation lowered the worst nearest-ring |k| (one test)
v1.1 (git ea9f761)	Reza: "Make nearest-ring the default definition of decoupled"	"Decoupled" = nearest ring; footprint verdict still reported. **Disclosed:** the λ_max worst-pair cost term had never been implemented	—	—
v1.1 addendum (git 6ce049e)	Reza: implement the worst-pair term and compare	Smooth k^4 worst-pair term	Paired comparison (35 runs)	All 3 criteria met, so it became the default. Trade-off: the median |k| rose
v1.2 (git 738072d)	Reza: bloom should lengthen lobes, not pinch	"Outward" bloom: fixed base circle, lobes extend; β up to 3	V6 re-run over 0.02–3.0	V6 FAIL only below the (doubled) bloom floor
v1.3 (git 30d6fa1)	Reza: petal sharpness, with the same topology checks	Power-law lobe profile	**V7** topology under sharpness	**V7 FAIL** (run 4): extra paired crossings in the flat drawing
v1.4 (git 7d7b6bf)	Fix after the V7 failure	Angle-warp sharpness (keeps q(p−1) crossings); grid crossing and clearance search; stricter point-count rule	—	Run 5: V7 FAIL at 1 of 360 points (resolution); run 6 with a higher point ceiling: **V7 PASS**
v1.5 (git ff1a40e)	Reza: "a quick visualization that's reasonable"	Quick B1 view (Biot–Savart; head surface and slices); dB coupling plot	**V8** on-axis loop B1	V8 PASS
v1.6 (git 91c4df5 pre-registration; 5f60ace implementation; f120e53 results)	Reza: symmetric arrays, better placement, controllable bloom and size, face zones,	Face exclusion zones (eyes, mouth); ring mode (mirror-symmetric, per-ring size/bloom/sharpness); multi-start; Voronoi adjacency;	**V9** symmetry, **V10** break placement, **V11** capacitor value, **V12** zones, **V13** reproducibility, **V14** B1	Development runs: **V14 FAIL** (midpoint rule, about 1%), fixed with exact segments; **V9 FAIL** (current-direction sign

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	independent B1 check, “let’s do the major update”	regularity score; capacitor breaks and first-pass values; save/load; **exact-segment B1**	vs Magpylib; comparison **C2** ring vs free	convention), fixed. Official: V7–V14 PASS. **C2: ring failed all 3 criteria, so free stays the default**
v1.7 (this version)	Reza: “implement the full capacitance modelling, including choosing the capacitors and also evaluating the effect of the capacitor in the decoupling and the B1 field” (head-loaded, Python, no MATLAB)	Full-wave co-simulation pipeline: openEMS (Python), lumped ports at the capacitor breaks, three-layer head phantom, circuit co-simulation that picks capacitor values with preamp decoupling, then capacitor effects on coupling and B1	V15–V20 and V15b (Design.md §7)	V15 FAIL (raw FDTD resonance −10%: thin-wire effective radius ≈ 0.17 × grid cell, confirmed across 3 grids); other results in the validation report
v1.8 (git 979b1fa pre-registration; implementation and results in the following commit), 2026-10-21	Reza: “add the height dimension to these knots, such that with one parameter, I could allow the knots to rise higher (total thickness of the whole knot)”; chose “outward only”	Total-thickness parameter T: h = max(h_min, T/2), whole knot raised by h − h_min so the lowest wire stays where it was (T = 0 = v1.7 exactly); app slider + side view; shared T in arrays. **Found while planning:** the self-inductance offset was not perpendicular to non-flat wires; fixed	**V21** thickness/topology, **V22** thick-knot B1 vs Magpylib, **V23** tilted-circle self-L	Development run: V21, V22, V23 PASS; V1, V2, V4, V5, V8, V11 re-run, still PASS. The v1.7 self-L method gave +11% at 80° tilt (V23, information)
v1.10 (git e2763ae pre-registration; bc374d0 implementation), 2026-10-21	Reza: is the output ready for COMSOL? “add the COMSOL export … first”	COMSOL export files from the same case JSON as openEMS (shared `sim/wirecut.py`); MPh build/solve/extract script (UNVALIDATED, written without COMSOL; dry-run tested); `cosim.py --solver comsol`	**V25** COMSOL vs openEMS	NOT EVALUABLE (no COMSOL here)
v1.9 (git 8fa1bf7 pre-registration; fb79dee, 2c4af52, cbbf6db implementation and fixes), 2026-10-21	Invalid pair simulation: the two coils’ wires touched where they cross (Design §4 lift never implemented). Reza	Crossover bumps: higher-numbered coil lifts outward over lower ones with a smooth spatial bump; coupling on bumped wires; V9 on the pre-bump layout; local grid refinement at crossovers, port-	**V24** crossover bumps	Development: 4 pair simulation cases PASS; **4 arrays FAIL** (bumps cap at 10 mm in dense knot arrays — open design problem). Taper width TUNED 6 → 10 mm on

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	chose “local bumps”	terminal snapping, minimum-cell guard; inter-element pre-run gate		the pair cases (bend limit). One aborted run (sliver grid cells, killed after 3 min)
v1.7 audit (night)	Reza: full audit in 4 steps	Brainstorm → independent review (Fable) → 4 area audits (Fable) → fixes. Main fixes: A1 clearance bug (lift too small for p ≥ 3 knots); M1 V15–V20 now in the report; C1 B1 plane offset; D2 pair-worker race; B1/B2 ring-mode report; licences; docs	PH3 (signed B1, post-hoc)	See `DECISIONS.md` “Audit actions”

Persistent open items (all versions)

- **Placeholders:** wire spec, helmet dimensions, face-zone positions, decoupling threshold, and the tissue properties at 3T.
- **V3:** cannot be evaluated until Roemer et al. 1990 is read.
- **V6:** fails below the wire-dependent bloom floor, a physical limit shown in the app.
- **No bench validation of anything yet:** every EM result is exploratory.